

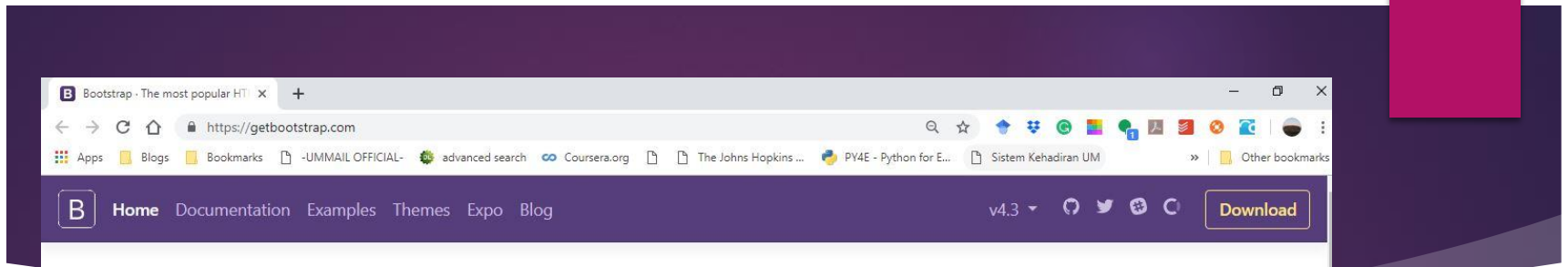


Introduction to Bootstrap

WIF2003 WEB PROGRAMMING

Objectives

- ▶ Define Bootstrap and explain why use it
- ▶ Include Bootstrap locally and by using a CDN
- ▶ Using common Bootstrap components like navs and buttons
- ▶ Building a layout using the Bootstrap grid system



Bootstrap

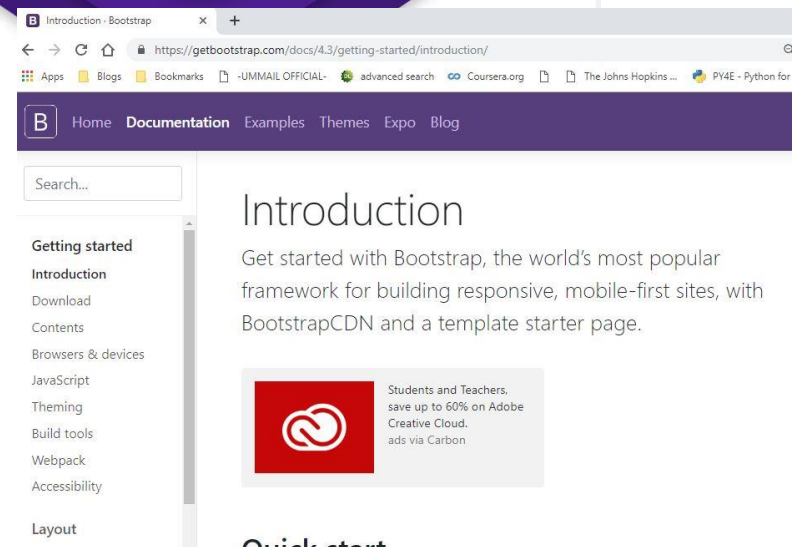
Build responsive, mobile-first projects on the web with the world's most popular front-end component library.

Bootstrap is an open source toolkit for developing with HTML, CSS, and JS. Quickly prototype your ideas or build your entire app with our Sass variables and mixins, responsive grid system, extensive prebuilt components, and powerful plugins built on jQuery.

[Get started](#)

[Download](#)

<https://getbootstrap.com/>



Overview of Bootstrap

A popular **front-end component library** for developing responsive, mobile first projects on the web

Bootstrap is an **open source toolkit** for developing with **HTML, CSS, and JS**

Help us quickly build responsive websites

Overview of Bootstrap (cont.)

Has good documentation and a lots of good examples

Like *Lego pieces* that we can put the pieces together and add them to our websites

Support the **latest, stable releases** of all major browsers and platforms.

Overview of Bootstrap (cont.)

Components

- Give us access to a bunch of pre-built components that we can incorporate into our own websites.

Grid System

- Come with a grid system, which helps us construct our own custom, responsive layouts.

JavaScript

- Powerful JavaScript plugins.

Bootstrap links

- ▶ Current version: v5.3
 - ▶ <https://getbootstrap.com/>
- ▶ Documentation - Getting started:
 - ▶ <https://getbootstrap.com/docs/5.3/getting-started/introduction/>
- ▶ Bootstrap Codepen Demo:
 - ▶ <https://codepen.io/team/bootstrap/pen/qBamdLj>

Bootstrap CSS

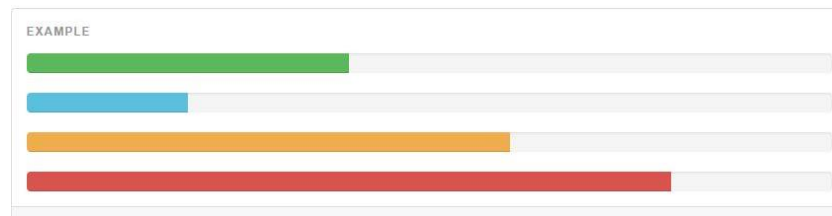
- ▶ Bootstrap is just a single CSS file, a single JavaScript file
 - ▶ <https://getbootstrap.com/docs/5.1/getting-started/contents/>
- ▶ There are different types of styles that we can use them for free
 - ▶ E.g. Tables, Buttons, Forms, Images, Navbars

Bootstrap Components

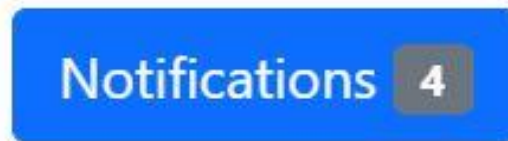
- ▶ **Reusable** components (bigger pieces) built to provide button group, carousel, dropdowns, navigation, alerts, and much more.
- ▶ **Plug and play** component that we can add into different parts of our web application
- ▶ One of the most popular aspects of Bootstrap and many websites use Bootstrap components

Bootstrap Components

► Examples:



- Progress: Provide up-to-date feedback on the progress of a workflow or action.



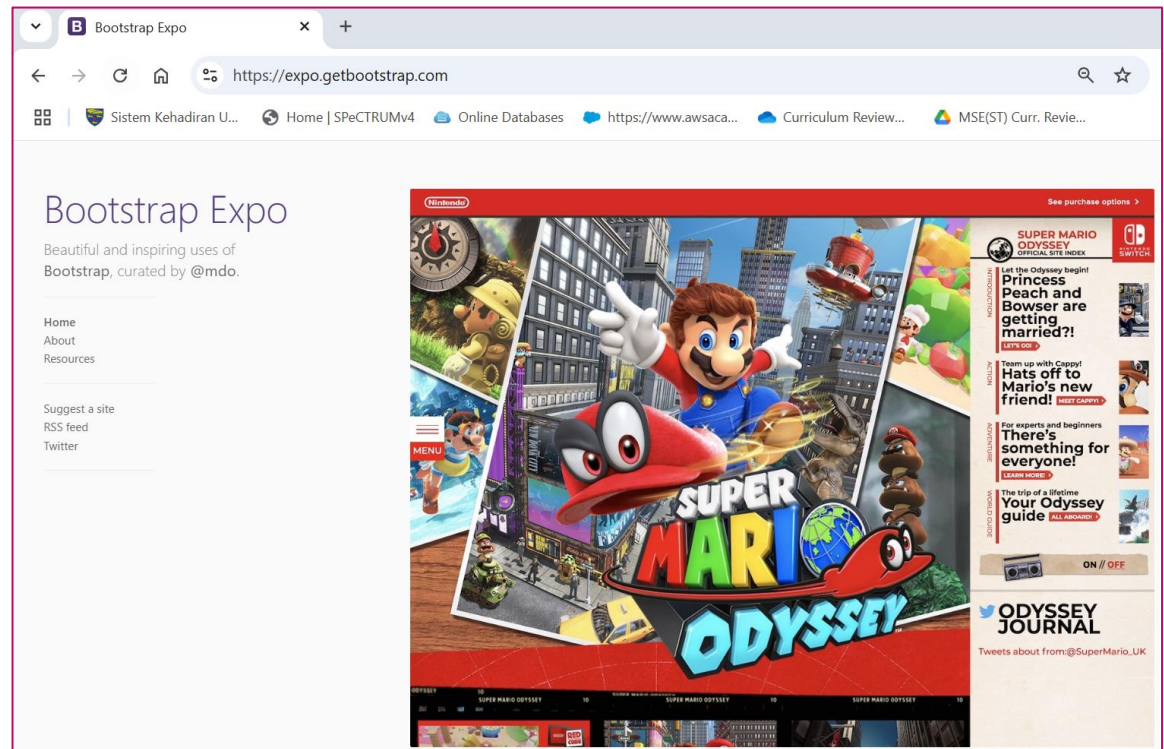
- Badges: Can be used as part of links or buttons to provide a counter.

Do all the websites that use Bootstrap look the same?

- ▶ The answer is 'Yes' and 'No'
- ▶ **No:** if a company or a project that uses Bootstrap well & use the main components differently
 - ▶ e.g., change the colors, change the fonts, change the hover effects
- ▶ **Yes:** if the websites use Bootstrap without changing anything at all

Bootstrap Expo

- Visit:
<http://expo.getbootstrap.com/>
- Showcase beautiful and inspiring uses of Bootstrap



Bootstrap Expo

- ▶ Explore projects that used Bootstrap
- ▶ Figure out what are the **core components** that many websites are using and **why** they decide to use Bootstrap



Option 1: Download and Include Bootstrap Locally

Download Bootstrap

- ▶ Download links for Bootstrap v5.3
 - ▶ <https://getbootstrap.com/docs/5.3/getting-started/download/>
 - ▶ Click the “**Download**” button
- ▶ Download Bootstrap to get the compiled CSS and JavaScript, source code, or include it with your favorite package managers like npm
- ▶ Easily drop them into your project

Add Bootstrap

- ▶ Unzip the file, you can see two folders:
 - ▶ **css**: Compiled and minified CSS bundles
 - ▶ **js**: Compiled and minified JavaScript plugins
- ▶ In CSS folder, the main css file is **bootstrap.css**, it has all the basic Bootstrap components
- ▶ **Add** the **bootstrap.css** file into the **same folder** that store the html files of our web applications
- ▶ Now, we can use the classes in the bootstrap.css file in our web applications

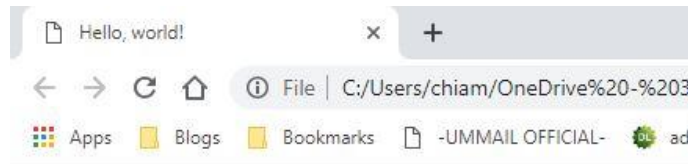
Include bootstrap.css in html

- Include the bootstrap.css using the **<link>** tag (same as how we include other external styles in the html)

```
<!DOCTYPE html>
<html>
<head>
<title>Bootstrap Demo</title>
<link rel="stylesheet" type="text/css" href="bootstrap.css">
</head>
<body>
    <h1>Bootstrap Basics </h1>
</body>
</html>
```

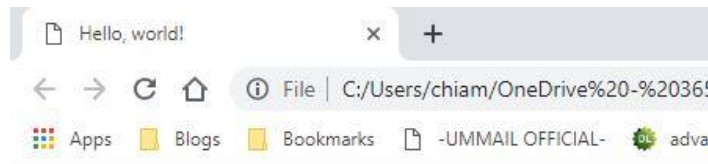
Before & After Using Bootstrap

- ▶ **Before** include bootstrap.css:



Hello, world!

- ▶ **After** include bootstrap.css:



Hello, world!



Option 2: Include Bootstrap
by using a CDN

Bootstrap CDN

- ▶ Link to external Bootstrap files that are hosted online
- ▶ Copy the Bootstrap CDN link from <https://getbootstrap.com/docs/5.3/getting-started/introduction/#cdn-links>
- ▶ Include the using the **<link>** tag (same as how we include other external styles in the html)

Include CDN links

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Bootstrap demo</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMHjY6hW+ALEwIH"
crossorigin="anonymous">
  </head>
  <body>
    <h1>Hello, world!</h1>
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz"
crossorigin="anonymous"></script>
  </body>
</html>
```

Start with template/examples

- ▶ Begin working with a minimal Bootstrap document
- ▶ Or modify the examples:
 - ▶ <https://getbootstrap.com/docs/5.3/examples>
 - ▶ Quickly get a project started with any of the examples ranging from using parts of the framework to custom components and layouts

Layout

- ▶ Version 5.3:
 - ▶ <https://getbootstrap.com/docs/5.3/layout>
- ▶ Components and options for laying out your Bootstrap project, including wrapping containers, a powerful grid system, a flexible media object, and responsive utility classes.
- ▶ Bootstrap requires a containing element to **wrap site contents** and **house our grid system**.
- ▶ You may choose one of two containers to use in your projects.

Layout: Containers

- ▶ Containers are the **most basic layout element** in Bootstrap and are required when using our default grid system.
- ▶ Choose from **a responsive, fixed-width** container (meaning its max-width changes at each breakpoint) or **fluid-width** (meaning it's 100% wide all the time).
- ▶ Example:
 - ▶ <https://getbootstrap.com/docs/5.3/examples/grid/#containers>

Add Container <div> in html

Use .container to set a max-width at each responsive breakpoint.

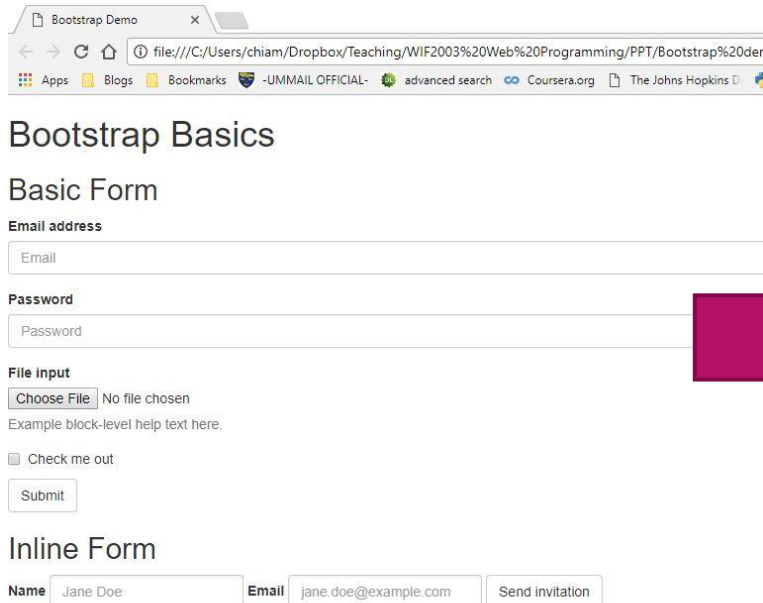
```
<body>
<!-- content goes in the body -->
<div class="container">
<!-- Content here -->
</div>
</body>
```

Use .container-fluid for a full width (100%) container, spanning the entire width of your viewport.

```
<body>
<!-- content goes in the body -->
<div class="container-fluid">
<!-- Content here -->
</div>
</body>
```

Container - Example

Before using `.container`
(no space)



Bootstrap Demo

file:///C:/Users/chiam/Dropbox/Teaching/WIF2003%20Web%20Programming/PPT/Bootstrap%20der

Apps Blogs Bookmarks -UMMAIL OFFICIAL- advanced search Coursera.org The Johns Hopkins D

Bootstrap Basics

Basic Form

Email address

Password

File input

No file chosen

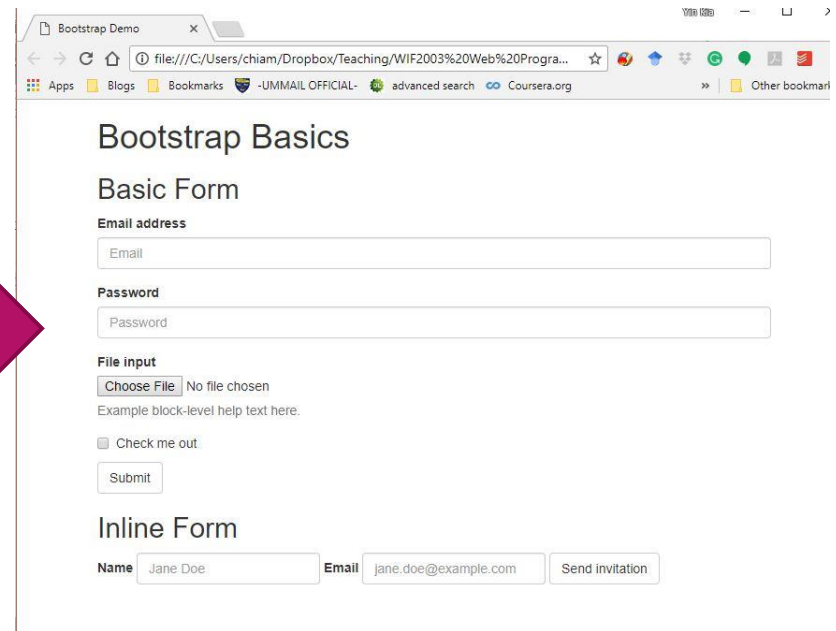
Example block-level help text here.

☐ Check me out

Inline Form

Name Email

After using `.container`



Bootstrap Demo

file:///C:/Users/chiam/Dropbox/Teaching/WIF2003%20Web%20Progra...

Apps Blogs Bookmarks -UMMAIL OFFICIAL- advanced search Coursera.org >> Other bookmark

Bootstrap Basics

Basic Form

Email address

Password

File input

No file chosen

Example block-level help text here.

☐ Check me out

Inline Form

Name Email



Using common Bootstrap components

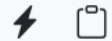
Buttons

- ▶ Version 5.3:
 - ▶ <https://getbootstrap.com/docs/5.3/components/buttons/>
- ▶ Use the button classes on an `<a>`, `<button>`, or `<input>` element.
- ▶ Use Bootstrap's **custom button styles** for actions in forms, dialogs, and more with support for multiple sizes, states, and more.

Buttons - Example



HTML



```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>

<button type="button" class="btn btn-link">Link</button>
```

Navigation:

Navs & tabs, Navbar

- ▶ **Navs & tabs:** Share general navigation component's **markup and styles**, from the base `.nav` class to the active and disabled states
- ▶ **NavBar:** Powerful, responsive **navigation header**; include support for branding, navigation, and more
- ▶ Version 5.3:
 - ▶ <https://getbootstrap.com/docs/5.3/components/navs-tabs/>
 - ▶ <https://getbootstrap.com/docs/5.3/components/navbar/>

Navs & tabs - Example

Active

Link

Link

Disabled

HTML

```
<ul class="nav nav-pills">
  <li class="nav-item">
    <a class="nav-link active" aria-current="page" href="#">Active</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Link</a>
  </li>
  <li class="nav-item">
    <a class="nav-link disabled" aria-disabled="true">Disabled</a>
  </li>
</ul>
```



Navbar - Example

Navbar Home Features Pricing About

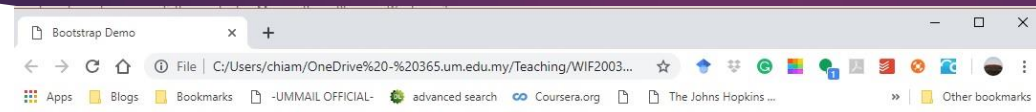
Navbar Home Features Pricing About

Navbar Home Features Pricing About

Forms

- ▶ Individual form controls automatically receive some global styling.
- ▶ All textual `<input>`, `<textarea>`, and `<select>` elements with `.form-control` are set to `width: 100%`; by default.
- ▶ Wrap labels and controls in `.form-group` for optimum spacing.
- ▶ Version 5.3:
<https://getbootstrap.com/docs/5.3/forms/overview/>

Forms - Example



Bootstrap Basics

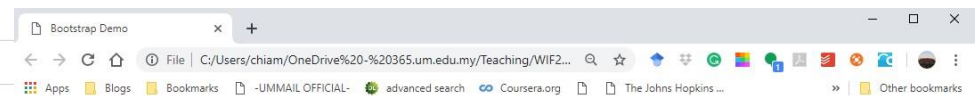
Basic Form

Email address

We'll never share your email with anyone else.

Password

☐ Check me out



Bootstrap Basics

Form Controls

Email address

Example select

Example multiple select

Example textarea

Override the bootstrap.css style (Screenshots)



Override the colors





Building a layout using the Bootstrap grid system

Grid System

- ▶ Bootstrap's grid system uses a series of **containers, rows, and columns** to **layout** and **align** content.
- ▶ It's built with **flexbox** and is fully responsive.
- ▶ Reference link:
 - ▶ Version 5.3:
<https://getbootstrap.com/docs/5.3/layout/grid/>

Overview of Grid System

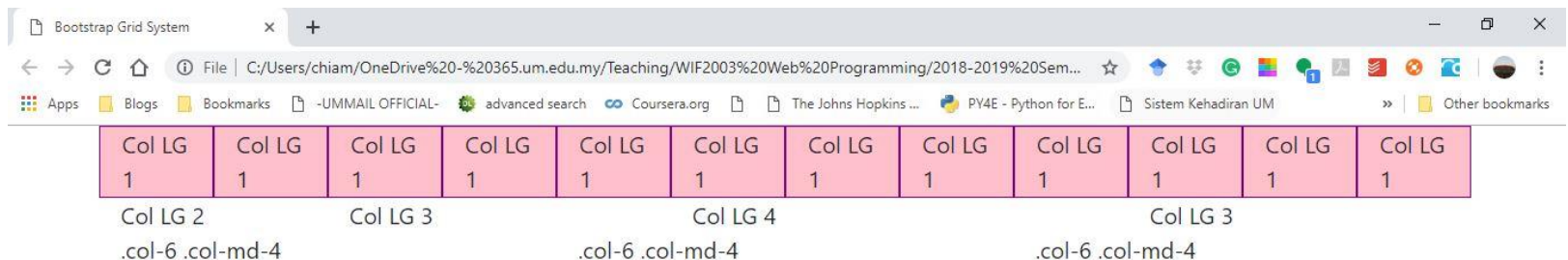
- ▶ Bootstrap includes a **responsive, mobile first fluid** grid system that appropriately scales up to 12 columns as the device or viewport size increases.
- ▶ It includes **predefined classes** for **easy layout options**, as well as powerful mixins for generating more semantic layouts.

How it works?

- ▶ Bootstrap's grid system uses a series of **containers**, **rows**, and **columns** to layout and align content.
- ▶ It's built with **flexbox** and is fully responsive.
 - ▶ The Flexbox Layout (Flexible Box) module aims at providing a more efficient way to lay out, align and distribute space among items in a container, even when their **size is unknown and/or dynamic** (thus the word "flex").

Grid System – Rows and Columns

- ▶ Rows must be placed within a `.container` (fixed-width) or `.container-fluid` (full-width) for proper alignment and padding.
- ▶ Use rows to create horizontal groups of columns.
- ▶ Scale up to 12 columns



The screenshot shows a web browser window titled "Bootstrap Grid System". The address bar displays a file path: `C:/Users/chiam/OneDrive%20-%20365.um.edu.my/Teaching/WIF2003%20Web%20Programming/2018-2019%20Sem...`. The browser's bookmark bar includes links to "Apps", "Blogs", "Bookmarks", "-UMMAIL OFFICIAL-", "advanced search", "Coursera.org", "The Johns Hopkins ...", "PY4E - Python for E...", and "Sistem Kehadiran UM".

Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1	Col LG 1
Col LG 2 .col-6 .col-md-4		Col LG 3		Col LG 4 .col-6 .col-md-4				Col LG 3 .col-6 .col-md-4			

Grid System – Example 1 (2 Rows and 12 Columns) Part 1

```
<body>
<div class="container">
  <div class="row">
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
    <div class="col col-lg-1 pink">Col LG 1</div>
  </div>
```

lg = large column, lg-1 = each column takes up 1 unit wide

Grid System – Example 1 (2 Rows and 12 Columns) Part 2

this column takes up 2 units wide

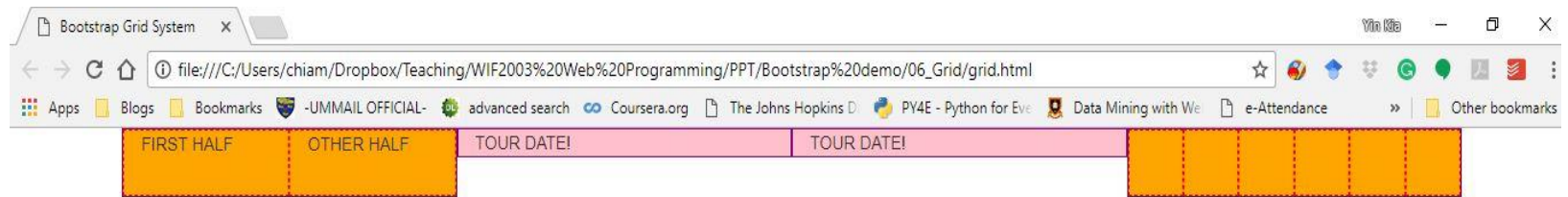
```
<div class ="row">
  <div class="col col-lg-2">Col LG 2</div>
  <div class="col col-lg-3">Col LG 3</div>
  <div class="col col-lg-4">Col LG 4</div>
  <div class="col col-lg-3">Col LG 3</div>
</div>
<div class="row">
  <div class="col-6 col-md-4">.col-md-4</div>
  <div class="col-6 col-md-4">.col-md-4</div>
  <div class="col-6 col-md-4">.col-md-4</div>
</div>
</div>
</body>
</html>
```

Columns start at 50% wide on mobile and bump up to 33.3% wide on desktop

Grid System – Example 2

```
<div class="container">
  <div class="row">
    <div class="col-md-3 col-sm-6 pink">
      <div class="row">
        <div class="col-lg-6 orange">FIRST HALF</div>
        <div class="col-lg-6 orange">OTHER HALF</div>
      </div>
    </div>
    <div class="col-md-3 col-sm-6 pink">TOUR DATE!</div>
    <div class="col-md-3 col-sm-6 pink">TOUR DATE!</div>
    <div class="col-md-3 col-sm-6 pink">
      <div class="row">
        <div class="col-lg-2 orange"></div>
        <div class="col-lg-2 orange"></div>
        <div class="col-lg-2 orange"></div>
        <div class="col-lg-2 orange"></div>
        <div class="col-lg-2 orange"></div>
        <div class="col-lg-2 orange"></div>
      </div>
    </div>
  </div>
</div>
</body>
</html>
```

Grid System – 2nd Example (Screenshot)





Override the
bootstrap.css style

Option 1: Override Bootstrap Styles using Embedded Styles

Example: Change the colors of button

```
<!DOCTYPE html>
<html>
  <head><!-- metadata goes in head -->
    <title>Bootstrap Demo</title>
    <link rel="stylesheet" type="text/css" href="bootstrap.css">
    <!-- override the bootstrap.css style -->
    <style type="text/css">
      .btn-primary{
        color: orange;
        background: purple;
      }
    </style>
  </head>
```

Learn More



Learn More

Option 1: Override Bootstrap Styles using Embedded Styles (cont.)

```
<body>
<h1>Bootstrap Basics</h1>
<h2>Override the bootstrap.css style</h2>
<button class="btn btn-lg btn-primary">Learn More</button>
<button type="button" class="btn btn-success btn-sm btn-lg
active">Click me</button>
<button type="button" class="btn btn-success btn-sm
disabled">Click me</button>
<a href="https://expo.getbootstrap.com/" class="btn btn-info btn-
lg">Bootstrap Expo</a>
</body>
</html>
```

Option 2: Override Bootstrap Styles using External Styles

Example: Sample Codes in the 'Project' folder

External CSS File: app.css

```
.lead {  
  font-size: 2em;  
}
```



A growing company who strive to deliver industry-standard quality within acceptable time and budget.

```
nav.navbar {  
  background-color: #6d6875;  
}
```



Summary

- ▶ This lecture introduced you to Bootstrap.
- ▶ You have learned how to use Bootstrap CSS common components to develop responsive web applications
- ▶ You have learned how to create page layouts using the Bootstrap containers and Grid System